Everseen Vision AI Hackathon Research Report: Intelligent Retail Shelf Auditing

Part 1 — Executive Research Brief

The challenge of maintaining on-shelf availability represents a critical operational bottleneck in modern retail environments, leading directly to lost revenue, diminished customer loyalty, and corrupted inventory forecasting data. Conventional automated shelf-auditing solutions frequently rely on single-frame, instantaneous object detection to identify stockouts. This baseline approach is inherently vulnerable to false positive alerts caused by temporary occlusions, such as shoppers standing in the aisle, dynamic lighting changes, or superficial product misplacements. When retail staff are bombarded with false alerts, alert fatigue sets in, rendering the technological investment effectively useless.

The most critical insight derived from this analysis is that single-frame visual perception is fundamentally insufficient for robust retail execution. A commercially viable system must fuse spatial awareness, specifically understanding the physical constraints of a planogram, with temporal verification, which involves monitoring anomalies over a sustained time window to calculate uncertainty. The strongest opportunity for differentiation within the hackathon constraints lies in moving beyond merely counting detected products toward a system that mathematically aligns the observed physical layout with the digital expectation, filtering out transient noise before any human intervention is requested.

The recommended concept, designated as the Temporal-Spatial Evidence Fusion System, directly addresses these limitations. It utilizes a lightweight edge-based object detector coupled with a modified Needleman-Wunsch sequence alignment algorithm to compare observed shelf states against digital planograms. Crucially, the architecture incorporates a temporal evidence buffer that tracks anomalies over multiple consecutive video frames, ensuring that a replenishment alert is only triggered when a stockout or misplacement is persistently observed with high statistical confidence. This concept deserves further development because it is highly original, mathematically defensible, computationally efficient for edge deployment, and capable of being prototyped convincingly within a strict 48-hour time limit without requiring massive data center infrastructure.

Part 2 — Retail Problem and Root-Cause Analysis

Understanding the operational problem requires separating theoretical inventory from physical reality. On-shelf availability measures the presence of a specific product in its exact designated location on the retail floor, ready for immediate customer purchase. This differs fundamentally from total store inventory, which reflects the stock recorded within the retailer's Warehouse Management System, including backroom reserves, transit stock, and items currently sitting in customer shopping carts.

Stockouts frequently occur despite inventory records showing available stock due to a pervasive phenomenon known as phantom inventory. Phantom inventory arises when physical stock is lost, stolen, damaged, or misplaced, yet the digital record remains artificially inflated. Because the enterprise software believes the product is currently in stock on the floor, automatic

replenishment orders to the distribution center are never generated. In these scenarios, shelf gaps, misplaced products, and phantom inventory create a vicious cycle: a misplaced product creates a false gap in one location while masking a true gap in another, leading to delayed replenishment and inaccurate planogram compliance data. Manual shelf audits attempt to bridge this gap but inherently fail to provide reliable coverage. Manual audits are slow, labor-intensive, and represent only a lagging indicator of shelf health; by the time an associate manually logs a stockout, multiple sales have already been lost.

When a shelf anomaly is detected, store employees must execute a complex operational decision tree. They must determine whether the item is physically in the backroom, if the planogram needs a total reset due to a promotional shift, or if a manual inventory adjustment is required to trigger a supply chain reorder. Therefore, a genuinely useful replenishment alert must transcend binary empty or full signals. It must specify the exact product category, its expected spatial location, the duration of the anomaly, and the operational priority based on the item's historical sales velocity. Retailers must prioritize between a high-selling empty shelf and a low-selling empty shelf because labor is finite; a missing facing of a top-selling item incurs a significantly higher opportunity cost than a misplaced niche product. The costs of false alarms are equally severe, resulting in unnecessary trips to the backroom that waste labor hours, which represent a massive expense in environments where hourly wages are escalating.

Practical constraints further complicate visual perception. Retail layouts feature changing promotional displays, reflective packaging, dynamic pricing labels, and highly variable lighting conditions. These constraints demand a system that does not simply map pixels to specific products, but rather understands the geometric structure of the shelf. By applying temporal filtering, retail workflows can be vastly improved without requiring a fully autonomous inventory system, empowering existing staff with targeted, high-confidence interventions rather than attempting to replace them entirely.

Failure Mode	Root Cause	Existing Detection Approach	Unresolved Limitation	Proposed Opportunity
Phantom Inventory	Misplaced stock or unrecorded shrink keeps WMS inventory artificially high.	Scheduled manual inventory counts and cyclical auditing.	Manual audits are infrequent, allowing phantom inventory to suppress automated reordering for days.	Continuous visual verification linked to planogram expectations to instantly flag persistent physical discrepancies.
Temporary Occlusion	Shoppers, carts, or staff blocking the camera's line of sight to the shelf.	Single-frame object detection counting visible products.	Generates false-positive stockout alerts because the system interprets hidden products as missing products.	Implement temporal verification buffers to require an anomaly to persist for a defined time window before alerting.
Planogram Non-Compliance	Staff stocking products in incorrect facings due to time	Periodic image capture via field representative mobile	Batch processing means errors are only caught post-execution,	Edge-based sequence alignment to dynamically detect

Failure Mode	Root Cause	Existing Detection Approach	Unresolved Limitation	Proposed Opportunity
	pressure or customer disruption.	applications.	leading to lost sales prior to correction.	misplacements in real-time without rigid spatial mapping.
Alert Fatigue	Legacy AI systems flagging every minor facing gap or empty slot regardless of business value.	Simple threshold-based alerting (e.g., alert if any slot is empty).	Does not account for product sales velocity or backroom availability, overwhelming staff with low-priority tasks.	Decision layer utilizing sales-velocity metadata to mathematically prioritize alerts, combined with uncertainty calibration.

Part 3 — Competitive Landscape and Prior Art

An exhaustive analysis of the competitive landscape reveals a mature but highly fragmented market for automated shelf monitoring, ranging from autonomous robotic fleets to fixed-camera edge solutions and crowd-sourced mobile applications.

A primary competitor in this space is Trax Retail, which utilizes computer vision and deep learning to digitize the shelf and maximize sales execution. The standard Trax pipeline relies heavily on field representatives capturing overlapping frames with mobile devices. The system produces a "realogram" to compare against the digital planogram, executing object detection, item recognition, and gap list delivery typically within 90 seconds. While highly accurate and commercially prioritized, this approach is fundamentally latent; it requires a human to initiate the capture, meaning it cannot provide continuous real-time stockout alerts.

Conversely, Simbe Robotics offers Tally 3.0, an autonomous inventory robot equipped with a dozen high-vision cameras, Intel 3D cameras, and HDR sensors. Tally is capable of capturing 15,000 to 30,000 products per hour, navigating autonomously through aisles. The system utilizes edge computing via NVIDIA GPUs to process data quickly. While exceptionally accurate and capable of fusing RFID data with optical data, the autonomous robotics approach requires significant initial capital expenditure, and the robots themselves can occasionally obstruct narrow aisles during peak shopping hours.

Focal Systems approaches the problem by deploying fixed edge-cameras to monitor shelves continuously. Focal utilizes deep learning to track on-shelf availability and triggers alerts directly to store staff. This continuous monitoring eliminates the latency of mobile apps, but the hardware configuration relies heavily on dense, localized camera networks, which can strain local network bandwidth if raw video is pushed to the cloud.

Everseen, the sponsor of this challenge, represents the state-of-the-art in edge AI deployment. Primarily known for their Evercheck loss prevention system at self-checkout, their Evershelf technology expands into aisle management by processing up to 6 petabytes of video data daily directly at the edge. Everseen explicitly emphasizes utilizing existing CCTV networks (minimum 2MP) to avoid rip-and-replace hardware costs, minimizing cloud bandwidth, and applying temporal logic to reduce false positive rates.

Academic research provides critical prior art for algorithmic planogram compliance. Foundational work by Yücel et al. demonstrates the use of a modified Needleman-Wunsch

algorithm for sequence alignment. Originating in bioinformatics for DNA alignment, this algorithm dynamically compares observed product sequences to reference planogram sequences, applying penalties for insertions (misplacements) and deletions (stockouts). Furthermore, the introduction of the Segment Anything Model 2 (SAM 2) by Meta provides zero-shot visual segmentation and streaming video memory tracking, presenting a new avenue for shelf gap segmentation without heavy per-item training. Retail-specific datasets, such as the Retail Product Checkout dataset (RPC) with its 83,739 images, and the GroZi-120 dataset containing real-world variable-quality imagery, provide the necessary foundation for training these perception layers.

Competitor / Approach	Organization	Actual Capability	Strengths	Limitations	Implications for Proposal
Mobile Realogram Matching	Trax Retail	90-second mobile audit turnaround.	High resolution, reliable commercial prioritization.	High latency, requires human-in-the-loop to initiate capture.	Solution must operate passively and continuously without manual triggering.
Autonomous Shelf Robot	Simbe (Tally 3.0)	15k-30k products scanned per hour.	Exceptional accuracy, hybrid cloud/edge execution, multi-sensor.	High initial CapEx, robots can obstruct retail aisles during operation.	Solution must leverage low-cost, fixed edge cameras to ensure accessible ROI.
Fixed Camera Networks	Focal Systems	Continuous fixed-angle shelf monitoring.	Continuous coverage, deep integration into task management.	Network bandwidth constraints and dense hardware requirements.	Emphasize lightweight edge processing and temporal filtering to reduce payload.
Edge AI Platform	Everseen	Massive edge processing across existing CCTV.	Operates on existing hardware, reduces false positives via local agents.	Relies on sophisticated proprietary edge infrastructure and models.	Prototype must demonstrate hardware-agnostic edge logic, differentiating via algorithmic reasoning.
Sequence Alignment	Yücel et al. (Academic)	1D spatial matching using Needleman-Wunsch.	Mathematically robust to minor perspective shifts and scale changes.	Lacks a temporal dimension for filtering transient occlusions in video.	Combine this spatial algorithm with a multi-frame temporal buffer.

The gap analysis reveals that while edge processing and continuous monitoring exist, the combination of algorithmic spatial alignment and temporal occlusion filtering remains

underutilized in lightweight deployments. Five credible opportunities for differentiation emerge:

1. **Temporal Uncertainty Filtering**: Utilizing a temporal evidence buffer that requires a gap to persist across multiple observations, effectively filtering out shoppers and temporary occlusions.
2. **Algorithmic Planogram Alignment**: Implementing sequence alignment to dynamically penalize insertions and deletions, mapping messy visual reality to structured digital plans without requiring pixel-perfect bounding box coordinates.
3. **Embedding-Based Zero-Shot Verification**: Utilizing contrastive visual embeddings to identify product mismatches without needing to train a custom classifier for every new stock-keeping unit introduced to a store.
4. **Business Value Prioritization**: Calculating a severity score for each detected anomaly based on the planogram's encoded sales velocity and the product's hierarchical importance.
5. **Multi-Modal Graceful Degradation**: Designing the architecture to gracefully degrade; if exact item recognition fails due to blur, the system falls back to bounding-box geometry and gap detection to alert staff of a structural anomaly.

The originality claim must be strictly bounded: the use of sequence alignment in retail is documented in prior academic literature. The exact proposed difference and core novelty lie in the combination of sequence alignment with a multi-frame temporal state machine to build an uncertainty-aware edge system.

Part 4 — Five Candidate Concepts

To avoid converging prematurely on a generic object detector, five distinct candidate concepts were formulated and evaluated against the hackathon constraints.

1. Pure Spatial Planogram Matcher (The Baseline) This concept functions as a highly rigid spatial reasoning system. It converts incoming shelf images into structured grids based on strict bounding box coordinates and calculates the Intersection over Union (IoU) against a perfectly scaled digital planogram grid. The key insight is that physical shelves should map exactly to digital layouts. The workflow involves detecting all objects, mapping their geometric centroids, and alerting on any spatial mismatch. The target user is a store manager seeking perfect compliance. While technically feasible for a 48-hour prototype, this system is exceptionally brittle. Any change in camera angle, focal length, or partial occlusion completely breaks the spatial mapping, leading to a high volume of false alerts.

2. Temporal-Spatial Evidence Fusion System (The Recommended Concept) This concept fuses algorithmic spatial awareness with a time-series state machine. The key insight is that a stockout in a physical store is a continuous state over time, not a single momentary snapshot. The workflow captures frames periodically, utilizes a lightweight detector to find generic products, extracts visual feature embeddings, and runs a 1-dimensional sequence alignment algorithm (Needleman-Wunsch) per shelf row. Crucially, it pushes these frame-level results into a temporal buffer. An alert is only dispatched to staff if the anomaly persists beyond a defined number of frames and the environmental uncertainty score remains low. It targets operational leads who prioritize actionable, high-confidence alerts. It is highly differentiated by its temporal buffer, scientifically defensible, and feasible for a 48-hour build using standard open-source libraries.

3. Zero-Shot Visual Anomaly Engine (The Fallback Concept) This concept abandons traditional object detection bounding boxes entirely, focusing on foundational segmentation. The

core insight relies on utilizing massive pre-trained vision models to identify out-of-distribution shelf spaces. The workflow feeds a shelf image into the Segment Anything 2 (SAM 2) model, which segments all distinct items and physical gaps based on visual prompts. It then extracts features using a Vision-Language Model to compare observed text embeddings against planogram text. While this eliminates the need for any retail-specific training data, the compute requirements are massive. The latency associated with running SAM 2 and large vision-language models on edge hardware makes it highly risky for a real-time alerting prototype, but it serves as an excellent fallback if traditional detection pipelines fail during development.

4. Cross-Modal Inventory Resolver This concept operates on the premise that vision alone is insufficient; true intelligence requires integration with Point of Sale (POS) and inventory systems. When the vision system detects a gap, it queries the backend inventory database. If the database registers zero stock, the system classifies it as a normal stockout and ignores it. If the database registers positive stock, the system flags a phantom inventory event for immediate investigation. While this concept delivers massive business value by directly attacking inventory shrinkage, it violates the hackathon constraint of relying solely on public datasets and ordinary resources, as it requires live integration with a proprietary, real-time enterprise database.

5. Edge-Optimized Lightweight Gap Segmenter This concept is designed specifically for extreme low-cost cameras and severely limited compute environments. The key insight is that identifying specific items is computationally expensive, but identifying empty dark spaces is cheap. The workflow applies classical computer vision techniques, such as morphological thresholding and contour detection, to find gaps in the physical shelf structure, ignoring product classification entirely. While highly deployable and easy to build in 48 hours, it provides very low business value compared to item-aware systems because it cannot detect misplaced items, cannot verify planogram compliance, and cannot prioritize alerts based on sales velocity.

Criteria	Weight	1. Spatial Matcher	2. Temporal Fusion (Rec)	3. Zero-Shot Anomaly	4. Cross-Modal	5. Gap Segmenter
Originality	20%	4/10	9/10	8/10	7/10	2/10
Technical Feasibility (48h)	20%	8/10	8/10	5/10	4/10	9/10
Retail Business Impact	20%	6/10	9/10	7/10	10/10	4/10
Prototype Demonstrability	15%	7/10	9/10	8/10	4/10	6/10
Scientific Defensibility	10%	5/10	9/10	7/10	6/10	3/10
Deployment Practicality	10%	6/10	8/10	3/10	5/10	9/10
Evaluation Plan Quality	5%	7/10	9/10	7/10	5/10	5/10
Weighted Total	**100%**	**5.80**	**8.70**	**6.55**	**6.15**	**5.35**

The evaluation matrix dictates that the Temporal-Spatial Evidence Fusion System (Concept 2) is the primary recommendation. It strikes the optimal balance between advanced algorithmic

differentiation and achievable implementation. The Zero-Shot Visual Anomaly Engine (Concept 3) is designated as the fallback architecture.

Part 5 — Recommended Product Specification

The finalized product is designated as **OptiShelf Temporal AI**, operating under the tagline: Uncertainty-Aware Shelf Auditing for the Real-World Retail Edge.
The fundamental problem statement this product addresses is that traditional retail vision systems generate excessive, costly false alerts by treating transient occlusions as confirmed stockouts. Furthermore, legacy systems struggle to map raw, noisy bounding boxes to strict digital planograms, leading to missed misplacement detections. The target users are Retail Operations Managers who require actionable, high-confidence supply chain signals, and Store Associates who rely on mobile task lists for their daily replenishment workflows.
The core functionality of OptiShelf lies in its ability to ingest video streams from fixed edge cameras, detect products in a class-agnostic manner, align those products geometrically against a digital planogram using sequence alignment, and pass identified anomalies into a temporal state machine. The unique value proposition is that OptiShelf eliminates the false out-of-stock problem caused by human foot traffic, thereby lowering alert fatigue, increasing associate compliance, and maximizing the return on investment for edge infrastructure.
The user journey begins when a store associate receives an automated alert on their mobile device or handheld terminal indicating a high-priority out-of-stock event. The alert specifies the item name, aisle, shelf level, and explicitly notes that the anomaly has persisted for greater than fifteen minutes, confirming it is not a temporary blockage. The notification includes a visual crop of the exact shelf gap alongside a diagram of the expected planogram overlay. The associate retrieves the product from the backroom, restocks the physical shelf, and taps a resolution button on their device. Upon the next camera sweep, OptiShelf visually verifies the physical restock, cross-references the state machine, and automatically clears the anomaly from the central store dashboard, closing the operational loop.

Part 6 — Technical Architecture

The OptiShelf system is designed as an event-driven edge pipeline, systematically transforming raw pixels into structured, verified business events without routing raw video payloads to centralized cloud servers.

```
flowchart TD
    %% Inputs
    V[Camera Feed / Video Frames] --> P
    M[Digital Planogram JSON] --> R
    S[Sales Velocity Metadata] --> D

    %% Perception Layer
    subgraph Perception Layer
        P[YOLOv9 Edge Detector] -->|Class-Agnostic Boxes| E[Embedding
Extractor]
        E -->|Feature Vectors| G[Geometric Shelf Row Segmenter]
    end
```

```
    %% Reasoning Layer
    subgraph Planogram Reasoning Layer
        G -->|Observed 1D Sequences| R[Needleman-Wunsch Aligner]
        R -->|Mismatch Flags| C[Confidence Calibrator]
    end

    %% Temporal Layer
    subgraph Temporal Verification Layer
        C -->|Frame Level Anomalies| T[Temporal State Machine]
        T -->|Buffer N Frames| V2[Verification Filter]
        V2 -->|Filters out transient occlusions| D2[Persistent
Anomalies]
    end

    %% Decision & Action Layer
    subgraph Decision & Action Layer
        D2 --> D[Priority Scoring Engine]
        D -->|Severity > Threshold| A[Alert Generator]
        A --> U[Store Associate Mobile App]
    end
```

The system initiates in the **Inputs** module. The architecture ingests standard RGB video frames, accompanied by a timestamp and a unique camera identifier. A structured JSON file represents the digital planogram, detailing the expected product sequences, moving from left to right, for each individual shelf row. Optional sales metadata is ingested to dictate the commercial priority of each specific item.

Within the **Perception Layer**, a lightweight object detector performs the initial localization. Instead of utilizing a massive multi-class classifier that must be retrained every time a new product is introduced, the detector identifies all objects in a class-agnostic manner, simply labeling them as generic products. Visual embeddings are then extracted from each localized bounding box using a lightweight convolutional neural network. Finally, these bounding boxes are clustered along the y-axis to dynamically segment the image into distinct, horizontal physical shelf rows, preparing the data for 1-dimensional analysis.

The **Planogram Reasoning Layer** introduces the core algorithmic differentiation. The digital planogram is loaded as a target sequence of objects and quantities. The detected items, sorted by their x-coordinates, form the observed sequence. These two sequences are aligned using the modified Needleman-Wunsch algorithm. Gap penalties are applied dynamically; deletions represent undetected items indicating a stockout, while insertions represent unexpected items indicating a misplacement. This mathematical alignment handles messy, non-linear deformations in perspective far better than strict geometric bounding box overlaps.

The **Temporal Verification Layer** is responsible for managing uncertainty. Frame-level anomalies flagged by the reasoning layer are pushed into a temporal buffer, maintaining a state history over the last several observations. If a physical gap is detected, but the overall bounding-box count in that specific region fluctuates wildly from frame to frame, the system infers that a shopper or object is actively moving in front of the shelf. The uncertainty score rises, and the state machine suppresses any alert until the visual field stabilizes and the gap remains consistent over a pre-defined temporal threshold.

In the **Decision Layer**, anomalies that successfully survive the temporal filter are

mathematically scored. The priority is calculated as a function of the base severity of the event, multiplied by the sales velocity of the item, and adjusted by the visual confidence score. An unrecognized item inserted into a premium promotional slot generates an immediate investigation alert, while an empty slot for a low-selling item may be deferred to the next scheduled maintenance window. Finally, in the **Action Layer**, alerts are deduplicated to prevent spamming the staff, and a prioritized task list containing visual proof is transmitted to the end-user interface.

Part 7 — Model and Technology Selection

Selecting the appropriate models for edge-based retail analytics requires balancing inference speed against accuracy in highly cluttered environments.

Component	Baseline Approach	Improved Recommended Architecture	Justification
Product Localization	YOLOv5 (Medium)	YOLOv9 or RT-DETR	RT-DETR inherently handles intra-class variations and dense object packing better in retail environments. YOLOv9 offers superior real-time edge processing for small, dense objects.
Product Identification	Dense Multi-Class Classifier (e.g., 1,000 output nodes)	Class-Agnostic Detection + Visual Embeddings	Retail inventory changes constantly. An embedding approach utilizing cosine similarity requires no retraining when a new product is added; the system only needs the reference image database updated.
Gap Detection	Bounding box math (counting missing boxes)	SAM 2 (Segment Anything 2) Memory tracking	While basic box math is fast, SAM 2 provides zero-shot segmentation of irregular spaces and streaming memory across video frames. However, bounding box math combined with Needleman-Wunsch is recommended for the 48-hour prototype due to SAM 2's heavy compute limits.

Component	Baseline Approach	Improved Recommended Architecture	Justification
Planogram Matching	Simple IoU Thresholding	Modified Needleman-Wunsch Alignment	Handles messy, non-linear deformations in camera perspective by treating shelf layouts as 1D sequences with dynamic penalties for insertions and deletions.

The recommended baseline involves training a standard YOLO model to detect a small handful of specific grocery classes. The system counts the boxes in an image, and if the count drops below the expected planogram threshold, an alert is triggered. This baseline will fail in a live demonstration because it cannot detect order misplacements (e.g., swapping two identical products), it fails completely when an untrained product is introduced, and it triggers false alerts the moment a judge waves their hand in front of the camera.

The improved architecture leverages a pre-trained, class-agnostic YOLO detector to find generic products. It passes the resulting image crops to a lightweight network to generate feature embedding vectors. It then utilizes cosine similarity to match observed embeddings to expected planogram reference embeddings. Crucially, it applies the Needleman-Wunsch algorithm to penalize out-of-order sequences, mapping the observed physical reality to the logical planogram structure. Finally, a sliding time window ensures the anomaly persists for a minimum duration before acting. This improved approach is mathematically robust to new products, solves the occlusion problem via the temporal buffer, and provides exact planogram compliance metrics that baseline object detectors cannot achieve. Performance benchmarking on standard edge hardware (e.g., NVIDIA Jetson) will be required to guarantee frame-rate latency requirements are met.

Part 8 — Data and Experimental Plan

Developing a robust computer vision model requires vast amounts of high-quality, domain-specific data. Several publicly available datasets have been verified for retail applications.

The **Retail Product Checkout (RPC)** dataset contains 83,739 images and is exceptionally large-scale and fine-grained. It features varying clutter levels categorized as easy, medium, and hard, making it highly relevant for training embedding extractors to distinguish between visually similar packaging. It is licensed under CC BY-NC-SA 4.0, permitting non-commercial hackathon use. The **GroZi-120** dataset provides images of 120 grocery products captured in real-world environments, specifically video frames of actual shelves, which is crucial for testing model resilience against real-world lighting, blur, and perspective distortion. The **Freiburg Groceries Dataset** contains 5,000 images across 25 food classes and serves as an excellent baseline for classification testing. Finally, the **SKU-110K** dataset features extremely dense bounding box annotations for packed supermarket shelves, making it the ideal ground-truth source for fine-tuning the class-agnostic product localization model.

Because no single open-source dataset perfectly represents the complex interplay of temporal occlusions and strict planogram compliance, the team must construct a controlled,

supplementary evaluation dataset during the hackathon. The strategy involves staging a physical mock-shelf using standardized items. The team will capture continuous video clips representing specific test states.

The evaluation metrics are explicitly defined to measure both perception accuracy and operational decision-making. Perception metrics include localization mean Average Precision (mAP) for the bounding boxes, and Top-1 Identification Accuracy for matching the visual embedding to the product catalog. More importantly, the operational decision metrics include Stockout Precision and Recall (evaluating if real stockouts are identified and false stockouts ignored), the Transient False Alert Rate (measuring how many alerts are erroneously generated during an occlusion test), and the Time to Verified Alert (measuring the latency between the physical removal of an item and the generation of a high-confidence notification).

Part 9 — Working Prototype Blueprint

To execute this architecture within the 48-hour hackathon finale, the team must rely on a streamlined, Python-centric technology stack. The hardware baseline assumes standard GPU-accelerated laptops or accessible cloud instances.

The software stack relies on Python 3.10 and PyTorch for the inference framework. OpenCV handles rapid image preprocessing, while the Ultralytics API is utilized to deploy pre-trained YOLOv8/v9 models quickly for localization. The sentence-transformers library, utilizing a lightweight ResNet or CLIP variant, handles the image similarity embeddings. For the user interface, Streamlit provides the fastest path to a live dashboard, capable of displaying a side-by-side view of the live camera feed, the digital planogram, and a real-time event log. Data storage for the 48-hour prototype will rely on in-memory Python dictionaries and JSON files to entirely avoid the overhead of configuring a relational database.

The repository structure will be modular: an app.py root file for the interface, a models directory housing weights, and a core directory containing distinct scripts for perception, sequence alignment, temporal buffering, and decision scoring.

The 48-hour execution schedule is divided into rigorous operational blocks:

Time Block	Deliverable	Definition of Done	Likely Risks	Fallback Plan
Hours 1-8	Foundation & Data	Repo established, physical mock shelf staged, planogram.json authored.	Lighting conditions on the mock shelf are poor.	Rely on pre-recorded dataset videos instead of a live webcam.
Hours 9-16	Perception Layer	Pipeline draws accurate bounding boxes and labels them via embeddings.	Embedding similarity threshold is too sensitive.	Fallback to a basic classification model for the 5 mock items.
Hours 17-24	Planogram Alignment	Algorithm outputs a similarity score between observed shelf and JSON.	Y-axis coordinate clustering fails to segment shelf rows.	Hardcode the horizontal shelf row pixel boundaries.
Hours 25-32	Temporal Buffer	State machine delays output until anomaly persists	State logic becomes asynchronous and	Implement a simple rolling average of

Time Block	Deliverable	Definition of Done	Likely Risks	Fallback Plan
		for 10 frames.	crashes the video loop.	confidence scores instead.
Hours 33-40	UI & Integration	Live video plays in browser alongside the alert log.	Streamlit video processing induces heavy latency.	Process every 5th frame instead of full 30 FPS.
Hours 41-48	Testing & Pitch Prep	4 evaluation scenarios recorded and validated.	Live demo crashes due to memory leak.	Present pre-recorded screen captures of the successful runs.

Part 10 — Demonstration Strategy

The live demonstration is the critical moment of evaluation. The narrative must emphatically prove to the judges that the system is not a basic object detector, but a spatially and temporally aware reasoning engine capable of driving real store operations. The demonstration is structured around five controlled scenarios.

Scenario 1: Baseline Compliance (The Control) The demonstration begins with a camera focused on a perfectly stocked physical shelf. The Streamlit dashboard ingests the frame, compares it to the planogram, and displays a green status indicating 100% compliance. This establishes the system's spatial grounding and baseline accuracy for the judges.

Scenario 2: The Transient Occlusion (The Innovation) A team member steps directly in front of the camera, blocking 50% of the products on the shelf, lingers for three seconds, and walks away. A conventional system would instantly trigger a massive stockout alert. Instead, OptiShelf displays a yellow warning indicator reading "Visibility Obscured / Uncertainty High." Crucially, no restock alert is fired. Once the person leaves, the status smoothly returns to green. The judges learn that the temporal evidence fusion successfully prevents costly false-positive alerts.

Scenario 3: The Persistent Stockout (The Action) A product is visibly removed from the shelf. The system detects the physical gap, and the user interface displays "Analyzing..." as the temporal buffer cycles. After five seconds of sustained absence, a red, high-priority alert fires, confirming the stockout and detailing the exact item requiring replenishment. The judges learn that the system reliably registers true state changes but mandates temporal verification.

Scenario 4: The Misplacement (Spatial Reasoning) Two visually distinct items are swapped on the physical shelf. The total bounding box count remains identical, meaning a basic gap detector would notice nothing wrong. However, the Needleman-Wunsch aligner flags the sequence mismatch, firing an alert regarding a planogram violation and specifying the misplaced items. The judges learn that sequence alignment mathematically understands layout rules beyond mere detection.

Scenario 5: The Unknown Item An unfamiliar, un-programmed item (e.g., a coffee mug) is placed on the shelf. The class-agnostic detector finds the object, but the embedding extractor returns zero similarity to the product catalog. The system generates an "Unknown Insertion" alert, proving its resilience against out-of-distribution items without requiring complete model retraining.

Part 11 — Market and Business Case

The primary customer segments for this solution are enterprise supermarket chains, large-format convenience stores, and dedicated retail operations teams. The operational problem driving adoption is the massive, recurring labor cost associated with manual shelf audits, which can exceed $100,000 annually per location, combined with the $112 billion annual retail shrinkage problem across the industry.

A major barrier to enterprise adoption is the capital expenditure required to retrofit entire stores with dense camera networks or autonomous robotics. OptiShelf's edge-optimized architecture directly addresses this barrier by designing models small enough to run on existing 2MP CCTV infrastructure, severely reducing integration friction and deployment costs. The system does not attempt to replace existing Warehouse Management Systems; rather, it acts as a high-fidelity visual sensor layer that feeds verified, localized data via standard JSON webhooks into those existing platforms, complementing the digital inventory with physical reality.

To calculate Return on Investment (ROI), a retailer must evaluate a hypothetical framework based on their own internal metrics. If a 50,000 square foot grocery store suffers a 4% out-of-stock rate on its top 100 highest-velocity SKUs, identifying these stockouts just two hours earlier via automated alerts can prevent 15 abandoned purchases per day. Assuming an average basket value of $10 per missing item, the system recovers $150 per day, per store. This translates to roughly $54,000 in recovered revenue annually per location, which easily justifies the deployment of lightweight edge-compute nodes and a software-as-a-service licensing model.

In terms of privacy and responsible design, the system operates in environments heavily populated by shoppers. By utilizing a temporal architecture that processes raw video frames entirely at the edge, no video payload is transmitted to the cloud. The system deliberately excludes facial recognition or biometric tracking; humans are classified strictly as "occluding obstacles," and their pixel data is instantly discarded from memory. This data minimization strategy ensures strict compliance with global privacy regulations, such as GDPR, and protects the retailer from unnecessary liability while safeguarding consumer privacy.

Part 12 — Judge Challenge and Risk Register

A sophisticated hackathon judging panel will actively attempt to dismantle the proposal's assumptions. The team must proactively address these critical challenges.

1. **Existing Competitors**: *Why is your solution different from Trax or Everseen's existing products?* **Response**: Trax relies on latent, human-triggered mobile photos, which cannot provide continuous alerts. While Everseen deploys massive edge infrastructure, our specific algorithmic fusion of Needleman-Wunsch sequence alignment with a multi-frame state machine mathematically reduces the false positive rate on localized, low-power edge devices, providing a highly efficient alternative.
2. **True Technical Novelty**: *Did you invent the Needleman-Wunsch algorithm?* **Response**: No, Needleman-Wunsch is a foundational bioinformatics algorithm that has been adapted for planograms in recent academic literature. Our specific engineering novelty lies in wiring the geometric output of this sequence aligner into an uncertainty-aware temporal buffer to drive operational, priority-scored retail alerts.
3. **Planogram Data Availability**: *Stores notoriously have outdated planograms. How does your system function if the baseline JSON is incorrect?* **Response**: We acknowledge this weakness. If the reference planogram is strictly wrong, the system will generate false alerts. However, the embedding-based pipeline allows the system to operate in a

"discovery mode," automatically updating its own baseline realogram when a store manager explicitly verifies that a physical shelf is correct via the mobile interface.

4. **Misplacement vs. Shadows**: *How do you distinguish between an actual shelf gap and a black product packaging blending into the shelf shadows?* **Response**: Single-frame detectors fail completely here. Our temporal buffer leverages varying lighting as shoppers move past the aisle. If the spatial alignment expects a product but the visual embeddings yield low confidence due to shadows, the system triggers a low-priority "Verification" task for the next associate in the aisle, rather than a definitive "Stockout" alert.
5. **Compute Costs**: *Can this highly complex pipeline actually run on affordable edge hardware?* **Response**: Yes. By isolating heavy compute (the YOLO detector and Embedding extractor) only to frames that register initial pixel-level motion changes, and utilizing highly optimized 1D sequence math for the layout alignment, the pipeline minimizes continuous GPU duty cycles, making it suitable for deployment on standard edge nodes.
6. **Similar Looking Products**: *How does your system distinguish between visually identical variants, like Diet Cola and Regular Cola?* **Response**: The class-agnostic YOLO model only finds the bounding box. We rely entirely on high-resolution embedding similarity (e.g., CLIP) mapped to a reference library. If the cosine similarity margin is too close to call, the system flags an uncertainty event for human verification, prioritizing operational honesty over artificial confidence.
7. **Product Packaging Changes**: *What happens when a brand updates its packaging design?* **Response**: Because we utilize an embedding database rather than a rigid 1,000-class Convolutional Neural Network, the retailer only needs to upload one new reference image to the central catalog. The system instantly adapts without requiring expensive retraining cycles.
8. **False Positives**: *What happens if an employee stands in front of the shelf while actively restocking it?* **Response**: The temporal buffer directly mitigates this. The "Occlusion" logic suppresses all alerts until the shelf view remains entirely unobstructed for a continuous, defined number of seconds.
9. **Camera Placement**: *Supermarket aisles are narrow. A fixed camera cannot capture an entire shelf in a single frame.* **Response**: We acknowledge this physical constraint. While our prototype assumes a direct line-of-sight view, a full production deployment would require an image-stitching module utilizing homography estimation to construct a global panoramic shelf view before executing the sequence alignment.
10. **Temporal Verification**: *Doesn't implementing a temporal buffer introduce unacceptable latency into a "real-time" system?* **Response**: It introduces a deliberate 15-to-30 second latency. In retail store operations, a 30-second delay in generating a stockout task is operationally meaningless, but filtering out a single false alert saves five minutes of an employee's time. This operational trade-off is highly favorable.
11. **Performance on Unseen Products**: *How does the model react to a seasonal promotional item it has never encountered?* **Response**: The class-agnostic detector will still draw a bounding box around the physical object. The embedding will subsequently fail to match the digital planogram. The sequence aligner will accurately flag it as an "Unknown Insertion," alerting staff to investigate the discrepancy.
12. **Deployment at Scale**: *How do you practically scale this software to 1,000 individual stores?* **Response**: The system utilizes containerized edge deployment, leveraging Kubernetes on local store compute nodes. The cloud is strictly reserved for metadata routing and centralized alert dashboards, fundamentally solving the massive video

bandwidth bottleneck that plagues centralized vision systems.

13. **Ground-Truth Availability**: *You did not train your model on proprietary, real-world store CCTV. How valid is your prototype's performance?* **Response**: We utilized validated public datasets like GroZi-120 and RPC to verify the embedding extraction. The fundamental system architecture is logically sound; training on real CCTV data in production would merely fine-tune the perception weights, not alter the core temporal reasoning.
14. **Evaluation Validity**: *Are your evaluation metrics actually robust enough for enterprise deployment?* **Response**: We deliberately separate perception metrics (like mAP) from business metrics (like the False Alert Rate). A system with a marginally lower mAP but superior temporal filtering yields vastly better business metrics, which is the exact value proposition of OptiShelf.
15. **Commercial Adoption**: *Why would a store purchase this system if they already possess an expensive Warehouse Management System?* **Response**: WMS platforms rely heavily on checkout data and assume perfect physical operations. A WMS cannot digitally "see" phantom inventory, theft, or dropped items. OptiShelf acts as an independent visual verification layer that feeds verified ground-truth data into the WMS, correcting its inherent blind spots.

Part 13 — Initial Submission Recommendations

The initial proposal must be concise, punchy, and highly focused on the specific operational value of the innovation.
Title: OptiShelf: Temporal-Spatial Evidence Fusion for Resilient Retail Auditing **Tagline:** Eliminating alert fatigue through time-aware planogram sequence alignment.
The submission outline should flow as follows:

- **Problem Definition**: Explicitly highlight the limitations of generic object detection, specifically emphasizing alert fatigue, phantom inventory, and transient occlusions.
- **Proposed Solution**: Introduce the multi-stage architecture: Class-Agnostic Detection \rightarrow Visual Embeddings \rightarrow Needleman-Wunsch Planogram Alignment \rightarrow Temporal Evidence Buffer.
- **Innovation and Differentiation**: Clearly state that the scientific novelty lies in the *integration* of sequence alignment with temporal state machines to create an uncertainty-aware alert engine, rather than claiming to invent a new neural network.
- **Technical Architecture**: Include the detailed Mermaid diagram and firmly highlight the edge-compute feasibility.
- **Expected Store Actions**: Detail how alerts are mathematically prioritized by sales velocity and visual confidence, directly improving the store associate's workflow.
- **Market Relevance**: Point to industry leaders like Everseen and Simbe, emphasizing how OptiShelf provides a low-CapEx, high-accuracy approach leveraging existing camera infrastructure.
- **Evaluation Strategy**: Commit to a live, four-scenario physical mock-up demonstration to definitively prove temporal resilience against human foot traffic.

Part 14 — Final Decision Memo

To: Everseen Vision AI Hackathon Team (IIT Jodhpur) **From**: Lead AI Systems Architect

Subject: Final Development Directives for Prototype Execution

The multidisciplinary analysis concludes that the single recommended concept to pursue is the **Temporal-Spatial Evidence Fusion System**.

The Exact Innovation to Prioritize: You must prioritize the engineering of the temporal state machine and the implementation of the Needleman-Wunsch sequence alignment. Submitting a generic YOLO object detector will not win this hackathon; definitively proving that your system can actively suppress a false alert when a person walks in front of the shelf will capture the judges' attention.

The Minimum Viable Prototype: You must construct a live webcam feed focused on a physical desk simulating a shelf. Implement a YOLOv8 pre-trained model for identifying generic objects. Wire a backend Python script to manage the temporal state buffer, and project the final output to a Streamlit dashboard displaying the real-time compliance status.

What Must NOT Be Attempted: Do not attempt to train a massive multi-class instance segmentation model, such as SAM 2, from scratch within the 48-hour window. Furthermore, do not attempt to integrate a live, relational database; utilize static JSON files for planograms and pre-trained weights to eliminate configuration risks.

The Three Most Important Experiments:

1. *The Occlusion Test*: Verify that an object blocking the camera for three seconds does not trigger a stockout alert.
2. *The Misplacement Test*: Swap two physical items and verify that the sequence aligner catches the transposition without a drop in the overall bounding-box count.
3. *The Illumination Test*: Dim the room lights and verify that the uncertainty score rises, temporarily pausing alerts rather than misclassifying shadows as stockouts.

The Five Strongest Pieces of Evidence Supporting the Proposal:

1. Phantom inventory and manual audit delays cause compounding, measurable revenue loss in retail.
2. Academic literature validates the use of the Needleman-Wunsch algorithm for robust planogram alignment.
3. Pre-trained contrastive embedding networks eliminate the brittle dependency on per-SKU Optical Character Recognition.
4. Edge AI's capability to process video locally mitigates massive cloud bandwidth costs and privacy concerns.
5. Established retail datasets (RPC, GroZi-120) prove that fine-grained item recognition is feasible via embeddings.

The Next Ten Actions the Team Should Take:

1. Assign Daksh to set up the Streamlit UI and construct the temporal state machine logic in Python.
2. Assign Khushi to implement the YOLOv8/9 inference pipeline and integrate the Needleman-Wunsch alignment algorithm.
3. Construct a physical mock shelf utilizing five distinct, recognizable grocery items.
4. Author the planogram.json ground truth file mapping exactly to the mock shelf.
5. Capture test videos of the four defined demonstration scenarios to serve as a testing baseline.
6. Connect the perception output directly to the sequence aligner module.
7. Connect the output of the sequence aligner into the temporal buffer module.
8. Wire the finalized, filtered output stream into the Streamlit interface.
9. Rigorously rehearse the live "Shopper vs. Stockout" demonstration, ensuring timing is flawless.

10. Finalize the pitch deck, heavily anchoring the narrative on delivering "Actionable Intelligence over Raw Detection."

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